C loop3.c > ...

```c
#include <stdio.h>

int main() {
    float population = 100000;    // Initial population
    float rate = 0.10;            // 10% increase per year
    int year;

    printf("Year\tPopulation\n");
    printf("------------------\n");

    for (year = 1; year <= 10; year++) {
        population = population + (population * rate);  // Increase by 10%
        printf("%d\t%.0f\n", year, population);
    }

    return 0;
}

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Year    Population
------------------
1       110000
2       121000
3       133100
4       146410
5       161051
6       177156
7       194872
```

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Q. She population of a town is 1000 00. She population has increased steadily at the rate of 10% per year for the last 10 yrs.
WAP to determine the population at the end of each year in the last decad

```
#include <stdio.h>

int main() {
    float population = 100000;
    float rate = 0.10;
    int year;

  printf("Year \t Population \n");
  printf("-----------\n");

for (year = 1; year <= 10; year++) {
      population = population + (population * rate);
      printf("%d \t %.0f \n", year, population);
  }
return 0;
}
```

TEACHER'S SIGNATURE

```c
#include <stdio.h>

int main() {
  int n, i, j, num = 1;

  printf("Enter number of rows: ");
  scanf("%d", &n);

  for (i = 0; i < n; i++) {
    for (int space = 0; space < n - i; space++)
      printf(" ");

    num = 1;
    for (j = 0; j <= i; j++) {
      printf("%d ", num);
      num = num * (i - j) / (j + 1);
    }

    printf("\n");
  }

  return 0;
}
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc oc.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter number of rows: 3
   1
  1 1
 1 2 1
PS C:\Users\ACER\Desktop\itscbe>
```

b)

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

```
#include <stdio.h>
int main() {
    int n, i, j, num = 1;
printf("Enter number of rows:");
scanf("%d", &n);

for(i=0 ; i<n ; i++) {
   for(int space=0 ; space<n-i ; space++)
      printf(" ");

   num=1;
   for(j=0, j<=i ,j++) {
    printf("%d", num);
   num=num * (i-j) / (j+1);
  }
 printf("\n");
}
return 0;
}
```

```c
#include <stdio.h>

int main() {
  int i, j, k = 1, space;
  int rows = 3;

  for (i = 1; i <= rows; i++) {
    for (space = i; space < rows; space++) {
      printf(" ");
    }

    for (j = 1; j <= i; j++) {
      printf("%d ", k);
      k++;
    }

    printf("\n");
  }
  return 0;
}
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc lo.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
  1
 2 3
4 5 6
PS C:\Users\ACER\Desktop\itscbe>
```

Q. WAP to generate the following set of output.

a)
```
  1
 2 3
4 5 6
```

```
# include <stdio.h>

int main() {
  int i, j, k=1, space;
  int rows = 3;

for (i=1; i<=rows; i++) {
   for (space=i; space<rows; space++) {
        printf(" ");
  }

for ( j=1 ; j<= i ; j++) {
    printf ("%d ",k);
     k++;
 }
  printf("\n");
}
return 0;
}
```

```c
#include <stdio.h>

int main() {
  int num, i;

  // Input from user
  printf("Enter a number to print its multiplication table: ");
  scanf("%d", &num);

  // Print the multiplication table
  printf("\nMultiplication Table of %d:\n", num);
  for (i = 1; i <= 10; i++) {
    printf("%d * %d = %d\n", num, i, num * i);
  }

  return 0;
}
```

```
Multiplication Table of 43:
43 * 1 = 43
43 * 2 = 86
43 * 3 = 129
43 * 4 = 172
43 * 5 = 215
43 * 6 = 258
43 * 7 = 301
43 * 8 = 344
```

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```
printf("\nTotal negative no: %d", neg);
printf("\nTotal zeroes : %d\n", zero);

return0;
}
```

Q. WAP to print the multiplication table of the number entered by the user. It should be in the correct formatting.

Num * 1 = Num

→

```
#include <stdio.h>

int main() { → int num, i;
printf("Enter a number to print its multiplication table:");
scanf("%d, &num);

printf("\nMultiplication Table of %d : \n", num);
for( i=1 ; i<=10 ; i++) {
    printf("%d * %d = %d\n", num, i, num*i);
}
return 0
}
```

```c
#include <stdio.h>

int main() {
    int a, b, c, d;
    int limit;

    printf("Enter the limit (L): ");
    scanf("%d", &limit);

    printf("\nRamanujan Numbers up to limit %d:\n", limit);
    printf("-----------------------------------\n");

    // Check all combinations of cubes
    for (a = 1; a <= limit; a++) {
        for (b = a + 1; b <= limit; b++) {
            for (c = a; c <= limit; c++) {
                for (d = c + 1; d <= limit; d++) {
                    // Check if sums of cubes are equal
                    if ((a*a*a + b*b*b) == (c*c*c + d*d*d) &&
                        (a != c && a != d && b != c && b != d)) {
                        printf("%d = %d^3 + %d^3 = %d^3 + %d^3\n",
                            a*a*a + b*b*b, a, b, c, d);
                    }
                }
            }
        }
    }
```

```
PS C:\Users\ACER\Desktop\itscbe> gcc loop4.c
PS C:\Users\ACER\Desktop\itscbe> ./a.exe
Enter the limit (L): 3

Ramanujan Numbers up to limit 3:
-----------------------------------
PS C:\Users\ACER\Desktop\itscbe>
```

Q. Ramanujan Number is the smallest number that can be expressed as the sum of ~~the~~ two cubes in 2 diff ways. WAP to print all such numbers upto a reasonable limit.

→
```
#include <stdio.h>
int main () {
  int a, b, c, d;
  int limit;

 printf ("Enter the limit (L):");
 scanf (" %d", &limit);

 printf ("\nRamanujan Number Up to limit %d :n", Limits);
 printf ("- - - - - - - - \n");

for (a=1; a<= liimit; a++) {
 for (b=a+1; b<= limit; b++) {
  for (c=a; c<= limit; c++) {
   for (d=c +1; d<= limit; d++) {

  if (a*a*a + b*b*b)==(c*c*c+d*d*d) &&
    (a!=c && a!=d &&b !=c &&b != d)) {
   printf ("%d= %d^3+ %d^3 = %d^3 + %d^3\n", a*a*a
  + b*b*b , a, b, c, d);
} } } } }
 return 0;
}
```

```c
#include <stdio.h>

int main() {
  int num;
  int pos = 0, neg = 0, zero = 0;
  char choice;

  do {
    // Input a number
    printf("Enter a number: ");
    scanf("%d", &num);

    // Check if positive, negative, or zero
    if (num > 0)
      pos++;
    else if (num < 0)
      neg++;
    else
      zero++;

    // Ask user if they want to continue
    printf("Do you want to enter another number? (y/n): ");
    scanf(" %c", &choice);  // note the space before %c to consume newline

  } while (choice == 'y' || choice == 'Y');

  // Display the results
  printf("\nTotal Positive numbers: %d", pos);
```

```
Do you want to enter another number? (y/n): y
Enter a number: 2
Do you want to enter another number? (y/n): y
Enter a number: 0
Do you want to enter another number? (y/n): n

Total Positive numbers: 2
Total Negative numbers: 0
Total Zeroes: 1
PS C:\Users\ACER\Desktop\itscbe>
```

Loops

Q WAP to enter numbers till the user wants. At the end, it should display the count of positive, negative and zeroes entered.

```
int main() {
    char choice,

 do {
    printf ("Enter a number:");
    scanf ("%d", &num);

 if ( num>0)
     pos++;
else if (num<0)
     neg ++;
else
     zero ++;

printf (" Do you want to enter number? (y/n):");
scanf ("%c, &choice);

} while (choice =='y' || choice == 'Y');

printf ("\nTotal positive numbers : %d", pos);
```